

Measuring Computer System Performance

Performance in Computer Systems

Learning Objectives

Be able to...

- apply the “build-then-measure” approach to computer systems
- apply capacity measurements to computer systems
- define, calculate, and understand throughput and latency

Build-then-measure

- Performance is almost never what you expect from a whiteboard
- So, don't try to predict!
- Instead: build then measure
 - build an end-to-end system
 - measure the system to characterize performance using benchmarks
 - locate the constraint (where did the resources go?)
 - decide what to do next
 - repeat
- Capacity: the amount of a resource that a system can supply
 - CPU cycles/second, memory bytes, etc.
 - Estimate these based upon physical properties of the system
- Useful work: the portion of that resource spent *directly* for users
 - Memory used to store user's data, cpu time spent for user-facing compute, etc.
 - Measure these with tools
- Overhead: the portion of the resource that does not directly advance user's work
 - Context switches, virtual memory overhead, protocol headers, etc.
- Efficiency: percentage of (useful work / total consumption)